

General Transit Feed Specification Extensions for Eligibilities and Capabilities

End-of-Project Webinar
August 25, 2021



Project Background

June 2020: One of 17 projects selected for funding through the Federal Transit Administration Mobility for All Pilot Program, which aims to:

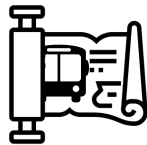
“improve mobility options through employing innovative coordination of transportation strategies and building partnerships to enhance mobility and access to vital community services for older adults, individuals with disabilities, and people of low income”

Thank You!

The project has been a tremendous success thanks to:

- Federal Transit Administration
- Grant administration from the National Aging and Disability Transportation Center (NADTC), a partnership of Easterseals, Inc. and the National Association of Area Agencies on Aging
- GTFS-eligibilities and -capabilities Expert Panel and Technical Working Group
- Support and guidance from ODOT's Karyn Criswell, Administrator, Public Transportation Division; Jennifer Sellers, Passenger Rail and Multimodal Network Manager; and the recently retired Matthew Barnes
- Project consultants, Full Path Transit Technology





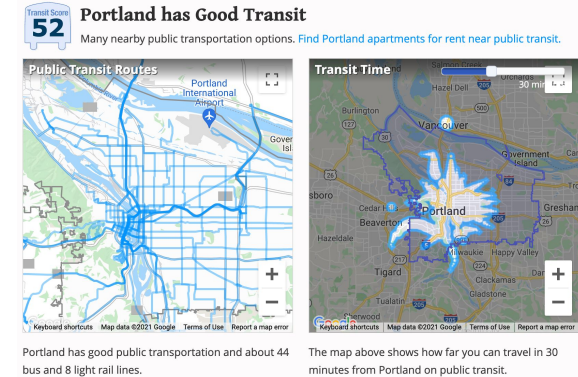
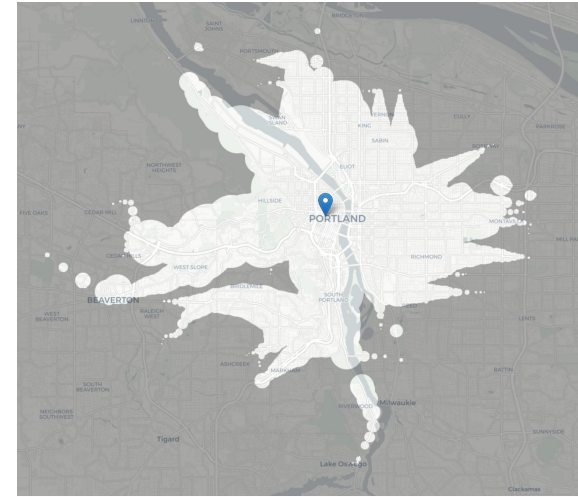
GTFS — An Overview

- 2006: *Google* Transit Feed Specification released
 - Static data only
 - For general public fixed route services only
- 2009: Renamed to the **General** Transit Feed Specification
- 2011: GTFS-Realtime extension released
 - Trip updates
 - Service alerts
 - Vehicle positions
- 2013: GTFS-Flex extension proposed to describe non-fixed-route transit
- 2019: MobilityData is formed and becomes the governance body for GTFS
- Today: The transit.land GTFS database lists feeds from ~5,300 operators in more than 60 countries



More on GTFS

- What does the GTFS and any proposed extensions need to be successful?
 - Producers
 - Consumers
- The spec has given rise to an ecosystem of producers and consumers
 - For the public: Mapnificent, WalkScore, Rome2Rio
 - For planners: Remix, Conveyal Analysis
 - For agencies (building tools for riders): OpenTripPlanner





What's *Not* Described? Attributes of Specialized Transit

- Any services not available to the general public:
 - In the US: Services complying with the Americans with Disabilities Act (ADA)
 - Older adults
 - People with disabilities
 - Veterans
 - Medical transportation
- Higher levels of service
 - Additional human resources available to riders
 - Details of accommodations for people with mobility devices



The Trip Lifecycle, In Six Stages



Discover

“What exists for people in my overall group(s)?”

Review Options

“Among the existing options, which options will work for my situation?”

Apply or Register

“What do I need to do to establish eligibility for the service?”

Sign In

“How do I connect to the service as a registered user?”

Request or Book Service

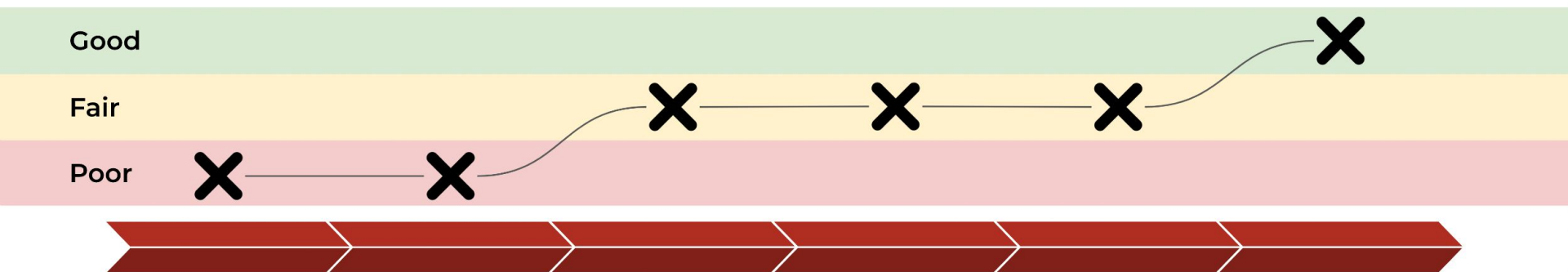
“What do I need to do to use the service?”

Service Provision

The actual transportation services and immediate lead-up



Sample User Experience Journey Map — Current



Discover

Very difficult to find all possible options.

Review Options

Very difficult to sift through and compare options.

Apply or Register

Varies greatly with service. Usually analog, the negative impact of which may not be great.

Sign In

Online profile for app- or web-based access is very rare. Negative impact of this may not be great.

Request or Book

Online options are rarely available. Negative impact of this may not be great.

Service Provision

Actual transportation services.

This Project: Draft 2 New Ways to Standardize Data

GTFS-eligibilities



GTFS-capabilities



The Two Extensions, Similarities



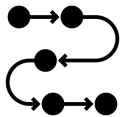
Are targeted to enhance the field's ability to describe specialized public transit services in a computer-readable format



Connect to the existing data model in GTFS and GTFS-flex (routes, trips, booking times, etc)



Will use a tabular (csv) file format



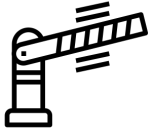
Result of project will be drafts. Both will need to pass through the same process as other GTFS extensions to become official (at least one producer and one consumer)



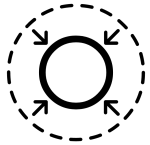
Contain personally identifiable information (PII) about riders

The Two Extensions, Contrasted

Eligibilities extension



Describes *constraints* to access for a given service.



Focused on a *narrowing* of service from the general public standard

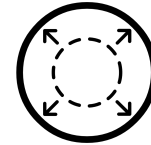


Needs to handle service-based or jurisdictional *rules* and *memberships*

Capabilities extension



Describes specifically what a service ***can do*** to respond to rider needs.

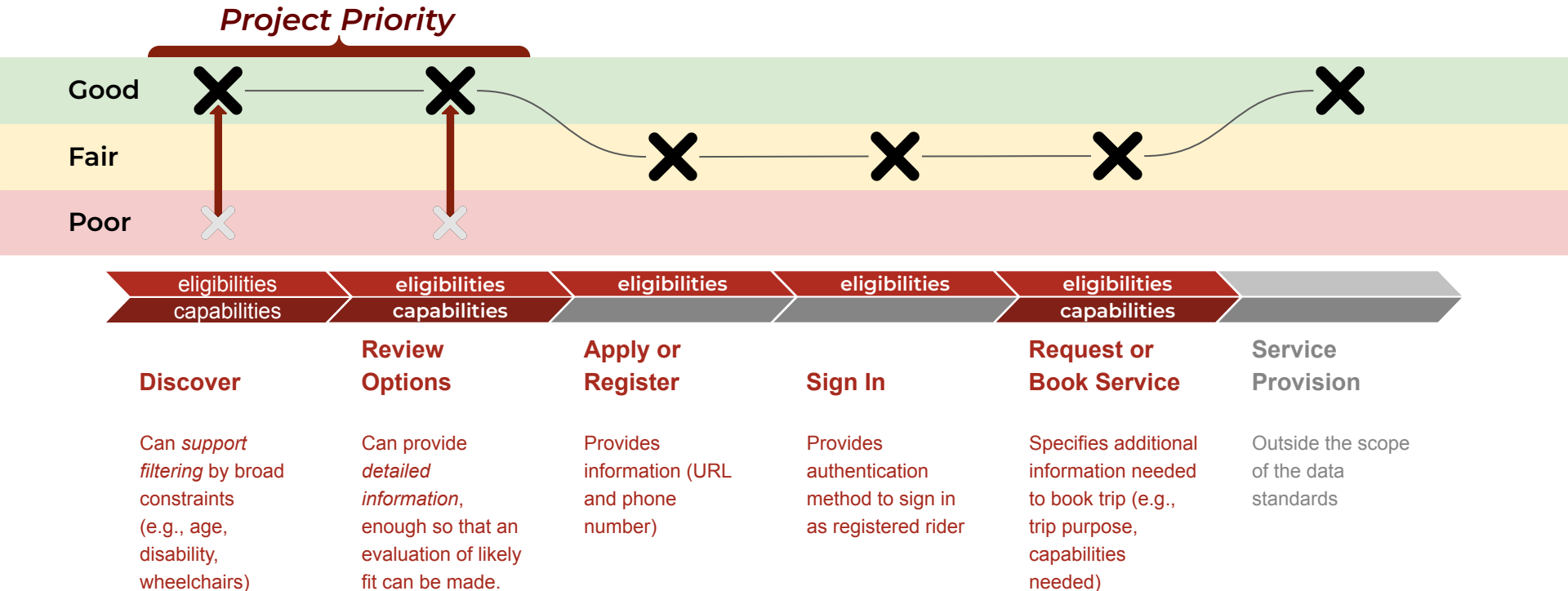


Focused on an *expansion* of service beyond those provided to ambulatory riders

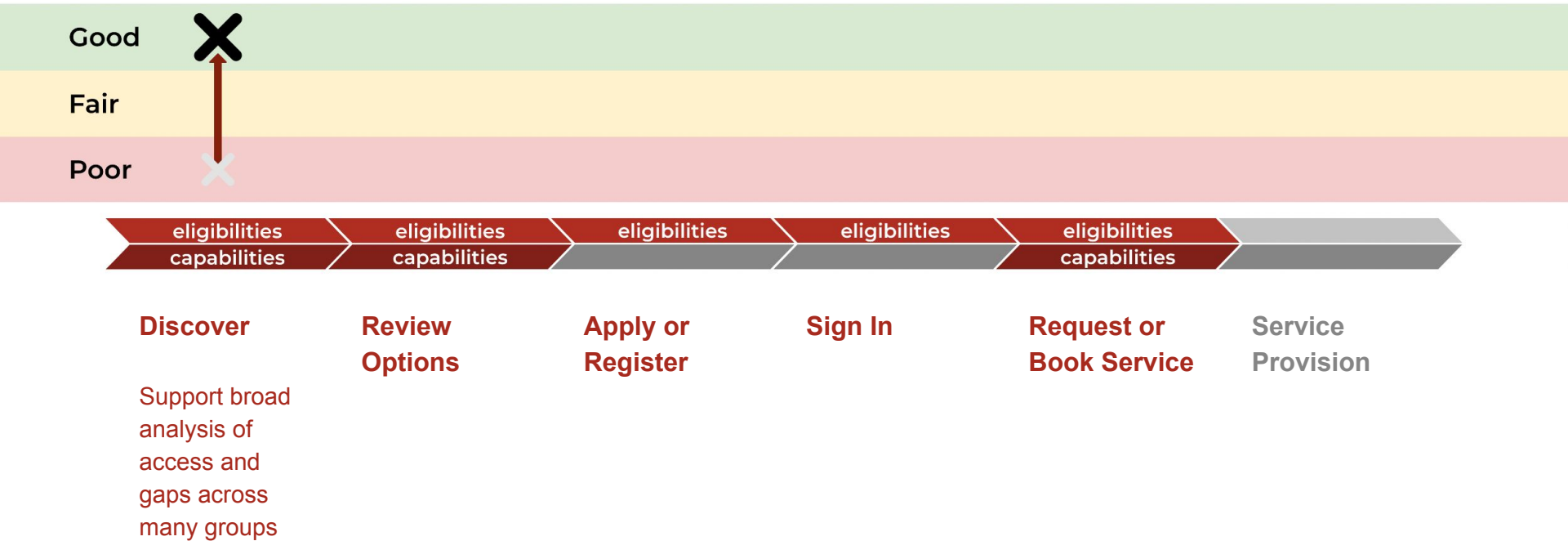


Needs to describe *spatial*, *mechanical*, or *human* resources

The Trip Lifecycle, Supported by Extensions, with Aspirational User Experience



Also, Allow For Better Tools For Planning, Policy, and Research





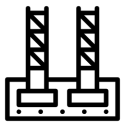
Design Goals

Overall

- Make the common cases easy
- Make more complex cases possible

For Eligibilities

- Make sure it's possible to capture the complexity of eligibilities as they exist today
- Provide a path to simplify and coordinate how eligibilities are managed at a regional level



What We Did — Built on Other Proposals

- For eligibilities:
 - GTFS-Fares (rider categories)
- For capabilities:
 - GTFS-VehicleCategories
 - GTFS-VehicleAllocations
 - GTFS-Seats



What We Did — We Added Service Attributes

- For eligibilities:
 - Minimum and maximum **age** constraint (already included in rider categories proposal)
 - Enumerated **basic other eligibility** constraint (riders with disabilities, other)
 - Enumerated allowed **trip purposes** (medical, shopping, education, work)
 - How eligibility is **checked** (at booking, at boarding)
 - How and where to **register** (URL, phone number)
 - Where to **appeal** an eligibility determination (URL, phone number)



Added Service Attributes, Part 2

- For human capabilities:
 - Enumerated **service levels** (curb to curb, door-to-door, door-through-door, hand-to-hand)
 - **Driver training** (has or has not received required training)
- For vehicle/mechanical capabilities
 - Presence and required use of **seat belts**
 - **Lift** presence, dimensions, and lifting capacity
 - **Ramp** presence, width
 - **Narrowest width** for accommodating the travel of a mobility device (including doorway)
 - Number of **stretchers** that can be accommodated



Introduced the Mobility Device Space

- Mobility device space attributes
 - Enumerated **securement methods** (no equipment, rider-operated equipment, driver-operated equipment)
 - Enumerated **location** (near the front door, near the rear door, near the rear of the vehicle, accessed by a separate door or lift)
 - Space **dimensions** (width, length)
 - Vehicle capacity **effect** (number of seats stowed, approximate number of standing riders displaced)
- Associate mobility device spaces with bookable seats (GTFS-seats proposal)
 - Describe how seats and mobility device spaces interact.



What We Did — Introduced the Concept of the **Uniform Resource Name (URN)**

- Type of Uniform Resource Identifier (URI), sibling to the Uniform Resource Locator (URL)
- Globally unique names, achieved through nested namespaces
- Clear structure rules allow for relatively easy creation of new names
- Intended to be a unique human-readable identifier, not a description (URN descriptions are also included in the proposal)
- Proposed to be used for three types of entities:
 - Eligibility
 - Trip purpose
 - Compliance
- Proposed format for GTFS:
urn:gtfs:{entity type}:{nested jurisdictions}:{organization}:{entity name}



URN Hypothetical Examples

- Eligibilities — three versions of veterans:
 - `urn:gtfs:eligibility:us:va:veteran`
 - `urn:gtfs:eligibility:us;oregon:odva:veteran`
 - `urn:gtfs:eligibility:us;oregon:ride.connection:2021.veteran`
- Trip purposes — recreation and three versions of medical
 - `urn:gtfs:trip.purpose:us;oregon;org:ride.connection:recreation;non-specific`
 - `urn:gtfs:trip.purpose:us;oregon;org:ride.connection:medical;non-specific`
 - `urn:gtfs:trip.purpose:us;oregon;org:ride.connection:medical;cancer.treatment`
 - `urn:gtfs:trip.purpose:us;oregon;org:ride.connection:medical;dialysis`
- Compliance — based on the “LEX” proposed standard for sources of law
 - `urn:gtfs:compliance:us:federal:session.law:1990-07-26;p1.101-336,104.stat.327`
(Americans with Disabilities Act)



What Do URNs Get Us?

- A chance to accurately name every current unique eligibility, trip purpose and law/regulation, in a mostly decentralized way, without risk of name collisions

Once you name something in a clear and shareable way, you can **do** stuff

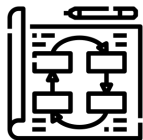
- A chance to **coordinate**, design, and share URNs across providers (e.g., at a regional or state level)
 - Example: regional reciprocity for ADA paratransit eligibility
- A chance to **link** eligibility to rider profiles
 - Identify a rider as a veteran, a resident of certain jurisdiction, etc
 - Facilitate discovery, booking, payment (e.g., reduced fare)

Result: increased legibility of specialized transportation services



What do these spec proposals help solve/not solve?

- They **do** help solve
 - Rider access (when used in tandem with apps)
 - Planning and evaluation (when used in tandem with the right tools)
- They **do not** solve
 - Issues with the service itself
 - Insufficient funding
 - Increased demand that may result from increased visibility
- Takeaway
 - Funding and service design matter



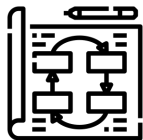
Expert Panel Members' Intentions and Plans

- **Data Producers:**

- Multiple providers state they are developing plans to create or provide data.
- *Trillium Solutions* (SaaS GTFS feed producer): “in the roadmapping stage of developing tools to build GTFS-Eligibilities/Capabilities data”

- **Data Consumers:**

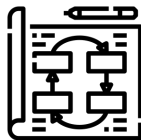
- *Interline* (SaaS transit data aggregation and management): “we can add support for GTFS-eligibilities/capabilities to Interline's "transitland-lib" processing libraries.”
- *Cambridge Systematics* (mobility software developer): “We hope to consume [the extensions] within 1-Click/OTP!”



Expert Panel Members' Intentions and Plans

- **Consultants/planners:**

- Carol Schweiger: "I plan to include them in the technology strategic planning and technology specification development that I do for transit/paratransit agencies [and] in the relevant training/capacity building courses that I develop for the USDOT's ITS Professional Capacity Building Program."

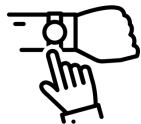


Expert Panel Members' Intentions and Plans

- **Proposal development and governance:**

- *MobilityData*: “we are definitely interested in using the components from GTFS-Eligibilities and GTFS-Capabilities that serve GOFS. That means MobilityData and the working group will aim to get these components pushed as official GTFS extensions.”

“As GTFS is an open spec, we can host GTFS-Eligibilities and GTFS-Capabilities in our resource center and include them in the roadmapping discussions with our members. We will be able to firm up in more detail which components we would like to use in GOFS as we continue building the spec over the coming two months.”



What does it all mean? It's time to...

- **Develop new relationships and expand existing ones** between transit and providers who have been outside the GTFS ecosystem
- **Identify** the problems that data in these new formats will help you solve
- **Engage riders** on what's most important to them
- **Make plans** to produce and/or consume in GTFS-flex, -eligibilities, and -capabilities.
- **Identify and address the barriers** to effectively managing the data for the long term
- **Support** the continued development and adoption of the extensions
- **Consider** becoming a member of MobilityData



Adoption Won't Happen On Its Own

Entities that are accountable to the public interest need to be present at the governance level. Involving public entities throughout the GTFS standards development process is important so that equity and interoperability can be given their proper weight.



Online Resources

- gtfspecs.org (hosted by Oregon DOT Public Transportation Division)
- gtfs.org
- mobilitydata.org
- github.com/ODOT-PTS
- github.com/full-path/gtfs-eligibilities (for the moment)
- github.com/full-path/gtfs-capabilities (for the moment)

Thank You!

Questions?

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